Lecture 7 Notes:
Aristotle's *Posterior Analytics*, Book II

Central Theme

Aristotle's *Posterior Analytics*, Book II gives an account of inquiry, definition, cause, and first principles.

Book I asked:

What makes a syllogism scientific demonstration?

Book II asks:

What are we looking for when we inquire scientifically?

$\boxed{\textit{Posterior Analytics}\text{, Book II} = \text{Aristotle's theory of inquiry, definition, and causal explanation}}$

Scientific inquiry is the search for the middle term: the cause that explains what a thing is or why a fact is so.

1. From Demonstration to Inquiry

Book I established what demonstration is: a syllogism productive of scientific knowledge.

Book II asks what we are seeking when we seek knowledge.

Book I: What is demonstration?

Book II: What do we seek in demonstration?

The answer is:

$\boxed{\text{We seek the cause, and the cause is the middle term.}}$

Teaching Point

Book II turns the theory of demonstration into a theory of inquiry.

Prior Analytics : middle term as connector

Posterior Analytics I : middle term as cause

Posterior Analytics II : middle term as object of inquiry

2. The Four Kinds of Inquiry

Aristotle says that there are four kinds of questions corresponding to the things we can know.

1. Whether a connection is a fact
2. Why the connection holds
3. Whether a thing exists
4. What the thing is

Questions About a Subject and Attribute

The first two questions concern whether a subject has an attribute and why it has that attribute.

Does the moon suffer eclipse?

Why does the moon suffer eclipse?

Questions About Existence and Essence

The second pair concerns whether a thing exists and what it is.

Does God exist?

What is God?

Whiteboard

$$\boxed{\text{That it is} \rightarrow \text{Why it is}}$$

$$\boxed{\text{Whether it is} \rightarrow \text{What it is}}$$

Teaching Point

Inquiry begins either with a fact to be explained or with a thing whose existence and nature are being sought.

$\boxed{\text{Scientific knowledge answers questions of fact, cause, existence, and essence.}}$

3. Every Inquiry Seeks the Middle

Aristotle says that all these questions ultimately concern the middle term.

When we ask:

Does the moon suffer eclipse?

we are really asking:

Is there a cause producing eclipse of the moon?

When we ask:

Why does the moon suffer eclipse?

we are asking:

What is that cause?

Whiteboard

Does the moon suffer eclipse? = Is there a middle?

Why does the moon suffer eclipse? = What is the middle?

$\boxed{\text{All scientific inquiry is inquiry into the middle term.}}$

Teaching Point

The middle term is no longer merely a logical bridge. It is the object of scientific search.

$\boxed{\text{To inquire scientifically is to search for the explanatory middle.}}$

4. The Middle Term Is the Cause

In the *Prior Analytics*, the middle term connected two extremes.

In the *Posterior Analytics*, the middle term explains why the extremes are connected.

$$\text{Subject} \rightarrow \text{Middle} \rightarrow \text{Predicate}$$

$$\text{Thing} \rightarrow \text{Cause} \rightarrow \text{Attribute}$$

Teaching Point

A scientific middle term is a cause.

$$\boxed{\text{The middle term answers the question “why?”}}$$

5. Essence and Cause

Aristotle closely links essence and cause.

For many things, to know what the thing is is also to know why the fact occurs.

Eclipse

Question:

What is eclipse?

Answer:

The privation of the moon's light by the interposition of the earth.

But this also answers:

Why does the moon suffer eclipse?

Answer:

Because the earth shuts out the light.

Teaching Point

For many scientific objects, the definition gives the cause.

$$\boxed{\text{For many things, to know what it is is to know why it is.}}$$

6. Definition and Demonstration

Book II raises a central problem:

> Can the same thing be known both by definition and by demonstration?

Definition and demonstration have different roles.

$$\text{Definition} \rightarrow \text{what something is}$$

$$\text{Demonstration} \rightarrow \text{that something belongs}$$

Example

Triangle has angles equal to two right angles.

This can be demonstrated.
But having angles equal to two right angles is not the definition of triangle.

Teaching Point

Definition and demonstration are related, but they are not identical.

$$\boxed{\text{Definition reveals essence; demonstration proves connection.}}$$

7. Not Everything Demonstrable Is Definable

Aristotle argues that not everything demonstrable can be defined.
A demonstrated conclusion often states that an attribute belongs to a subject.

Every triangle has angles equal to two right angles.

But the demonstrated attribute is not therefore the essence of the subject.

Teaching Point

A property may belong necessarily without being the definition of the subject.

$$\boxed{\text{Necessary property} \neq \text{essence}}$$

8. Not Everything Definable Is Demonstrable

Aristotle also argues that not everything definable can be demonstrated.

Definitions often function as starting points of demonstration. If every definition had to be demonstrated, demonstration would fall into infinite regress.

$$\text{definition} \rightarrow \text{starting point}$$

$$\text{demonstration} \rightarrow \text{proof from starting points}$$

Teaching Point

Some definitions are basic and indemonstrable.

$$\boxed{\text{Demonstration depends on definitions it does not itself prove.}}$$

9. Why Essence Cannot Simply Be Demonstrated

Aristotle considers whether there can be a syllogism of a thing's essence.

The difficulty is that a syllogism proves an attribute of a subject through a middle term. But essence is not merely an attribute added to a subject. It is what the subject is.

Problem

To prove the essence of man, one would need a middle term that already contains the essence of man.

$$\text{To prove essence, the middle must already contain essence.}$$

But then the proof assumes what it is supposed to prove.

$$\boxed{\text{The proof begs the question.}}$$

Teaching Point

Essence cannot be demonstrated as though it were an ordinary predicate.

$$\boxed{\text{Essence is not simply one more attribute attached to a subject.}}$$

10. Division Does Not Prove Essence

Aristotle criticizes definition by division.

Division proceeds by asking questions such as:

$$\text{Is man animal or inanimate?}$$

$$\text{Is animal terrestrial or aquatic?}$$

$$\text{Is terrestrial animal footed or footless?}$$

But each step is assumed, not inferred.

Teaching Point

Division may classify, but it does not demonstrate.

$$\boxed{\text{Division is useful for finding definitions, but it is not proof.}}$$

11. Why Division Is Not Demonstration

In a genuine demonstration, the conclusion follows necessarily from the premises.

In division, the definer asks for concessions and then assembles the formula. But the full definition does not follow with syllogistic necessity from the earlier choices.

Example

A divider may arrive at:

$$\text{man} = \text{animal, mortal, footed, biped, wingless}$$

But if asked why this whole formula is the essence of man, division alone does not supply a demonstration.

Teaching Point

Division can organize the search, but it cannot by itself prove that the definition is complete.

$$\boxed{\text{Division needs guidance from essence, not merely from classification.}}$$

12. Hypothetical Definitions and Begging the Question

Aristotle considers whether essence can be proved hypothetically.

One might say:

Suppose the definable form consists of certain peculiar attributes.

Suppose these are the only such attributes.

$\therefore$ this is the definition.

But Aristotle replies that the definable form has already been assumed in the premises.

Teaching Point

A definition cannot be smuggled into the premises and then presented as a conclusion.

If the proof assumes the definable form, it does not prove it.

13. Definition Does Not Prove Existence

A definition does not by itself prove that the thing defined exists.

Example

We may understand the phrase:

goat-stag

But understanding the phrase does not prove that goat-stags exist.

Likewise, one might understand a definition of circle without that definition alone proving the existence of circles.

Teaching Point

Meaning is not existence.

To know what a name means is not yet to know that the thing exists.

14. Starting Again: A Positive Account

After raising difficulties, Aristotle begins again.

He argues that to know a thing's essential nature is connected with knowing the cause of its existence.

If a thing has a cause distinct from itself, that cause can be exhibited through demonstration.

$$\text{Essence is not demonstrated directly.}$$

But:

$$\text{Essence can be revealed through a demonstration that gives the cause.}$$

Teaching Point

Demonstration does not prove essence directly, but it can exhibit essence through cause.

$$\boxed{\text{Causal demonstration reveals definable nature.}}$$

15. The Eclipse Example Revisited

Let:

$$A = \text{eclipse}$$

$$B = \text{earth's interposition}$$

$$C = \text{moon}$$

To ask whether the moon is eclipsed is to ask whether the defining condition exists.

$$\text{Is } B \text{ present?}$$

If the earth is interposed between sun and moon, then the moon suffers eclipse.

Definition

$$\text{Eclipse} = \text{privation of the moon's light by the earth's interposition}$$

Teaching Point

The cause that explains the fact also supplies the definition.

$$\boxed{\text{The cause of eclipse is contained in the definition of eclipse.}}$$

16. The Thunder Example

Aristotle gives another example: thunder.

Question:

What is thunder?

Answer:

The noise of fire being quenched in clouds.

Question:

Why does it thunder?

Answer:

Because fire is quenched in the clouds.

Whiteboard

Thunder = noise in clouds

Better:

$$\boxed{\text{Thunder} = \text{noise of fire being quenched in clouds}}$$

Teaching Point

Scientific definition is causal definition.

$$\boxed{\text{A real definition gives the reason why the thing is what it is.}}$$

17. Three Kinds of Definition

Aristotle distinguishes several senses of definition.

1. An indemonstrable statement of essence
2. A quasi-demonstration of essence in another form
3. The conclusion of a demonstration giving essence

Nominal and Causal Definition

Some definitions only state the meaning of a name.

$$\text{Nominal definition} = \text{what the name means}$$

Other definitions reveal the cause of a thing's existence.

$$\text{Causal definition} = \text{why the thing is what it is}$$

Teaching Point

Not all definitions are equally scientific.

$$\boxed{\text{The best definition is explanatory.}}$$

18. The Four Causes as Middle Terms

Aristotle says that we think we know scientifically when we know the cause.

He names four causes:

1. The definable form
2. The antecedent that necessitates a consequent
3. The efficient cause
4. The final cause

Each cause can function as the middle term of a proof.

$$\boxed{\text{The middle term can express different kinds of cause.}}$$

Teaching Point

Scientific explanation requires selecting the right kind of cause.

$$\boxed{\text{Different questions require different causal middles.}}$$

19. Formal Cause as Middle

Aristotle asks why the angle in a semicircle is a right angle.

The cause is that it is half of two right angles.

$$C = \text{angle in a semicircle}$$

$$B = \text{half of two right angles}$$

$$A = \text{right angle}$$

$$C \to B \to A$$

Teaching Point

The definable form can function as the explanatory middle.

$$\boxed{\text{The formal cause explains what the thing is.}}$$

20. Efficient Cause as Middle

Aristotle gives the example of the Persian War.
Question:

Why did war come upon the Athenians?

Answer:

Because they raided Sardis with the Eretrians.

Structure

$$C = \text{Athenians}$$

$$B = \text{unprovoked raiding}$$

$$A = \text{having war waged against them}$$

$$C \to B \to A$$

Teaching Point

The efficient cause explains the originating source of an event.

$$\boxed{\text{Efficient cause answers: What produced this?}}$$

21. Final Cause as Middle

Aristotle gives the example of walking after supper.

Question:

Why walk after supper?

Answer:

For the sake of health.

More specifically:

Walking after supper prevents food from rising.

Preventing food from rising contributes to health.

Structure

$$C = \text{walking after supper}$$

$$B = \text{non-regurgitation of food}$$

$$A = \text{health}$$

$$C \to B \to A$$

Teaching Point

The final cause explains the end for the sake of which something is done.

$$\boxed{\text{Final cause answers: For the sake of what?}}$$

22. Necessity and Final Causality

Aristotle notes that the same phenomenon may be explained both by necessity and by an end.

Example

Light passes through a lantern:

by necessity: because small particles pass through pores

for an end: so that we may avoid stumbling

Teaching Point

Aristotle allows layered explanations.

Necessity and purpose can coexist in explanation.

23. Nature, Necessity, and Chance

Aristotle distinguishes outcomes that happen:

always

for the most part

by chance

Natural processes often occur for an end, but they may also involve necessity.
By chance, however, nothing comes to be for an end in the strict explanatory sense.

Teaching Point

Scientific explanation must distinguish regular natural order from accidental coincidence.

Chance events do not have final causes as chance events.

24. Time and Causal Inference

Aristotle discusses cause and effect across time.
If the effect is present, the cause is present.
If the effect is past, the cause is past.
If the effect is future, the cause is future.

Eclipse

The moon was eclipsed because the earth intervened.

The moon is being eclipsed because the earth is intervening.

The moon will be eclipsed because the earth will intervene.

Teaching Point

The tense of the cause must match the tense of the explained fact.

Past effect requires past cause; present effect requires present cause; future effect requires future cause.

25. Circular Natural Processes

Aristotle notes that some natural processes occur in cycles.

Example

earth is moistened $\rightarrow$ exhalation rises $\rightarrow$ cloud forms $\rightarrow$ rain falls $\rightarrow$ earth is moistened

Teaching Point

In cyclic natural processes, causal explanation can move through reciprocal stages.

Nature sometimes displays circular processes of generation.

26. What Happens Always and What Happens For the Most Part

Some things happen always and in every case.

always

Other things happen as a general rule.

for the most part

Example

Not every man grows a beard, but it is the general rule.

For what happens for the most part, the middle term must also hold for the most part.

Teaching Point

The kind of conclusion depends on the kind of middle.

$$\boxed{\text{A for-the-most-part conclusion requires a for-the-most-part middle.}}$$

27. How to Obtain a Definition

Aristotle gives a method for finding definitions.

We select attributes that:

1. always belong to the thing,
2. are wider than the thing individually,
3. do not extend beyond the thing's genus,
4. together become coextensive with the thing.

Triad Example

A triad is:

number

odd

prime

Each attribute is wider than triad when taken separately, but together they define triad.

$$\boxed{\text{triad} = \text{number} + \text{odd} + \text{prime}}$$

Teaching Point

A definition gathers essential attributes until they are collectively coextensive with the thing.

$$\boxed{\text{Definition} = \text{essential synthesis with the same extension as the subject.}}$$

28. Division as a Useful Search Procedure

Although division is not demonstration, Aristotle gives it a positive role.

Division helps us:

1. admit only elements of the definable form
2. arrange them in the right order
3. omit none of them

Teaching Point

Division is not proof, but it is a disciplined way to search for definitions.

$$\boxed{\text{Division helps discover essence, though it does not demonstrate it.}}$$

29. The Right Order of Definition

The order of terms in a definition matters.

It is not the same to say:

animal, tame, biped

as to say:

biped, animal, tame

Definition should move from genus to successive differentiae in proper order.

$$\text{genus} \rightarrow \text{differentia} \rightarrow \text{lower differentia}$$

Teaching Point

A definition is not a heap of attributes. It is an ordered account of essence.

$$\boxed{\text{Definition requires structure, not mere accumulation.}}$$

30. The Three Rules for Definition by Division

When establishing a definition by division, Aristotle says we should keep three things in view.

1. Admit only elements in the definable form
2. Arrange them in the right order
3. Omit no essential elements

Teaching Point

A good definition is essential, ordered, and complete.

$$\boxed{\text{A definition must include all and only the essential parts in the right order.}}$$

31. From Particular Cases to Definition

Aristotle also recommends beginning from similar individuals and asking what they have in common.

Example: Pride

If we inquire into pride, we examine proud people such as:

Alcibiades

Achilles

Ajax

We ask what they have in common as proud.
Suppose we find:

intolerance of insult

Then we compare other cases, such as Lysander or Socrates, and ask whether there is one common formula.

Teaching Point

Definition is discovered by collecting common essential features.

$$\boxed{\text{Particulars} \rightarrow \text{common feature} \rightarrow \text{definition}}$$

32. Avoiding Equivocation in Definition

If inquiry produces not one formula but several, then the term may not name one thing. It may be equivocal.

Teaching Point

Definition requires unity. If there is no single formula, there may be no single thing being defined.

$$\boxed{\text{One definition requires one essence.}}$$

33. No Metaphors in Definition

Aristotle warns against metaphorical expressions in definition.

A definition should be clear, literal, and precise.

$$\text{metaphor} \neq \text{definition}$$

Teaching Point

Definitions must be perspicuous because they serve as starting points for knowledge.

$$\boxed{\text{Scientific definition requires clarity, not ornament.}}$$

34. Selecting Connections for Demonstration

To formulate what we wish to prove, Aristotle says that we should select analyses and divisions.

We begin with the common genus.

$$\text{animal}$$

Then we ask what belongs to every animal.

Next we move to proximate subgenera.

$$\text{bird}, \quad \text{horse}, \quad \text{human}$$

Then we ask what belongs essentially to these.

Whiteboard

$$\text{genus} \to \text{proximate subgenus} \to \text{species} \to \text{proper attributes}$$

Teaching Point

Scientific inquiry proceeds through ordered classification of genera and species.

$$\boxed{\text{To prove well, first classify well.}}$$

35. Common Characters Beyond Traditional Names

Aristotle warns that we should not confine ourselves to traditional class names.

We must collect any common character we observe.

Example

Horned animals may share:

possession of a third stomach

one row of teeth

The question then becomes:

To what species does the possession of horns belong?

Teaching Point

Scientific classification may require discovering new explanatory groupings.

Science follows real common characters, not merely inherited names.

36. Analogy in Scientific Selection

Sometimes there is no single identical name for a common structure.

Aristotle gives examples such as:

a squid's pounce

a fish's spine

an animal's bone

These may have analogous functions or structures, even if they lack one common name.

Teaching Point

Analogy can reveal common explanatory structure where language lacks a single term.

Scientific comparison may discover unity by analogy.

37. One Middle for Many Connections

Sometimes one middle explains several connections.

Examples

echo

reflection

rainbow

These differ specifically, but they are generically alike because they involve repercussion.

$\boxed{\text{common middle: repercussion}}$

Teaching Point

One causal structure may appear in different phenomena.

$\boxed{\text{Scientific explanation often discovers a shared cause across diverse cases.}}$

38. Subordinate Middles

Some explanations differ because one middle is subordinate to another.

Example

Question:

Why does the Nile rise toward the end of the month?

Answer:

Because the month is more stormy toward its close.

Question:

Why is the month more stormy toward its close?

Answer:

Because the moon is waning.

Teaching Point

Some causes explain through chains of subordinate middles.

$\boxed{\text{A cause may itself require a deeper cause.}}$

39. Cause and Effect: Must the Cause Be Present?

Aristotle asks whether, when the effect is present, the cause must also be present.

If we are dealing with universal and commensurate causal connections, cause and effect must reciprocate within the proper domain.

Example

If the cause of deciduousness is coagulation of sap, then:

$$\text{deciduousness} \leftrightarrow \text{coagulation of sap}$$

within the appropriate subject domain.

Teaching Point

Scientific cause must be proper and commensurate with the effect.

$$\boxed{\text{A true scientific cause explains all and only the relevant cases.}}$$

40. Cause Is Prior to Effect

Aristotle rejects the idea that an effect and its cause can simply explain each other in the same way.

The earth's interposition causes eclipse. Eclipse does not cause the earth's interposition.

$$\text{earth's interposition} \to \text{eclipse}$$

not:

$$\text{eclipse} \to \text{earth's interposition}$$

Teaching Point

Cause is prior to effect, even when cause and effect are convertible in a scientific domain.

$$\boxed{\text{Convertibility does not erase explanatory priority.}}$$

41. Can One Effect Have Many Causes?

Aristotle allows that the same effect may have different causes in different kinds of subjects.

Example

Longevity may have one cause in quadrupeds and another cause in birds.

$$\text{quadrupeds} \rightarrow \text{lack of bile}$$

$$\text{birds} \rightarrow \text{dry constitution}$$

Teaching Point

Plurality of causes is possible across kinds, but not within the same essential kind.

$$\boxed{\text{Different species may require different causes for similar effects.}}$$

42. The Proximate Cause Is the True Cause

When there are several middle terms, the true cause is the middle nearest to the species in which the property is manifested.

$$\text{remote cause} \rightarrow \text{general explanation}$$

$$\text{proximate cause} \rightarrow \text{proper explanation}$$

Teaching Point

The best explanation gives the nearest appropriate middle.

$$\boxed{\text{Scientific explanation seeks the proximate cause.}}$$

43. The Origin of First Principles

The final chapter asks how we come to know the primary immediate premises.

Aristotle rejects two views:

First principles are innate in a fully developed form.

First principles arise from nothing.

Instead, we have a natural capacity that begins with sense-perception.

Teaching Point

First principles are neither fully innate nor learned from higher principles.

$\boxed{\text{They arise from perception through a natural cognitive process.}}$

44. From Sense Perception to Science

Aristotle gives a developmental account.

sense perception → memory → experience → universal → art and science

Explanation

Sense-perception grasps particulars.

Repeated perception may persist as memory.

Many memories of the same thing become experience.

From experience, the universal becomes stabilized in the soul.

From this arise art and science.

Teaching Point

Science begins in perception but culminates in grasping universals.

$\boxed{\text{The universal emerges from repeated experience of particulars.}}$

45. The Battle-Rout Image

Aristotle compares the emergence of knowledge to a rout in battle being stopped.

First one soldier makes a stand, then another, and eventually the original formation is restored.

Likewise, among many perceptions, one stable universal first stands in the soul, then another, until knowledge becomes ordered.

Teaching Point

Knowledge arises when unstable perceptions settle into stable universals.

$\boxed{\text{Learning is the stabilization of the universal within experience.}}$

46. Induction and First Principles

Aristotle says that we come to know primary premises by induction.

$$\text{particulars} \rightarrow \text{induction} \rightarrow \text{universal} \rightarrow \text{first principles}$$

But induction here is not merely counting cases. It is the process through which the universal becomes present in the soul through repeated perception and memory.

Teaching Point

Induction prepares the mind to grasp first principles.

$$\boxed{\text{First principles are reached through induction from experience.}}$$

47. Intuition

Scientific knowledge depends on demonstration, but demonstration depends on primary immediate premises.

Therefore, scientific knowledge cannot itself be what grasps the first principles.

Aristotle concludes that the state that grasps first principles is intuition.

$$\text{intuition} \rightarrow \text{first principles}$$

$$\text{first principles} \rightarrow \text{demonstration}$$

$$\text{demonstration} \rightarrow \text{science}$$

Teaching Point

Intuition is the originative source of scientific knowledge.

$$\boxed{\text{Science begins from principles grasped by intuition.}}$$

48. The Architecture of Book II

Book II moves through the following arc:

$$\text{four questions} \rightarrow \text{middle term as cause} \rightarrow \text{definition vs. demonstration} \rightarrow \text{critique of proving essence}$$

$$\rightarrow \text{causal definition} \rightarrow \text{four causes} \rightarrow \text{methods of definition} \rightarrow \text{causal selection}$$

$\rightarrow$ proximate cause $\rightarrow$ origin of first principles $\rightarrow$ intuition

Teaching Point

Book II completes the *Posterior Analytics* by showing that scientific inquiry seeks causes, definitions, and first principles.

$$\boxed{\text{Inquiry is the search for the cause that makes knowledge possible.}}$$

49. Summary of Main Ideas

1. Scientific inquiry asks four questions: whether a fact is so, why it is so, whether a thing exists, and what it is.
2. All inquiry seeks the middle term.
3. The middle term is the cause.
4. Essence and cause are closely connected.
5. Definition and demonstration are related but not identical.
6. Essence cannot simply be demonstrated as if it were an ordinary predicate.
7. Division does not prove essence, but it helps discover definition.
8. Definition does not prove existence.
9. Demonstration can exhibit essence by revealing the cause.
10. Scientific definition is causal definition.
11. The four causes can function as middle terms.
12. Necessity and final causality can coexist in explanation.
13. Definitions are obtained by gathering essential attributes that become collectively co-extensive with the subject.
14. Scientific inquiry classifies genera, species, and proper attributes.
15. A single middle can sometimes explain many connections.
16. Scientific causes must be proper, prior, and proximate.
17. First principles arise from sense-perception, memory, experience, and induction.
18. Intuition grasps the first principles from which science demonstrates.

Final Framing

$\boxed{\textit{Posterior Analytics}\text{, Book II, shows that inquiry is the search for cause.}}$

Book I explains what demonstration must be. Book II explains what demonstration is looking for: the middle term that reveals the cause, the essence, and ultimately the first principles of science.

$\boxed{\text{Science demonstrates from principles, but intuition grasps the principles.}}$